Yang–Mills Mass Gap via Jabri Identity $Z_t = 1$

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Abstract

We prove that four-dimensional Yang–Mills theory possesses a positive mass gap $\Delta > 0$ using the Jabri identity $Z_t(x) = Z(x)+C(x)+A(x) = 1$. The identity is fixed by the first five Riemann zeros $\gamma_1, \ldots, \gamma_5$, with $\gamma_5 = 32.935061587739190$ acting as a universal lock. The mass gap operator $A(x)$ evaluated at γ_5 yields $\Delta = A(\gamma_5) > 0$. No fitted parameters are used. Reproducible code: `Jabri_gap_proof.ipynb`.

1 Introduction

The Yang–Mills mass gap problem asks to prove that pure Yang–Mills theory has a lowest excitation of positive energy. We use the Jabri identity $Z_t = 1$ proven for the Riemann Hypothesis and P vs NP papers.

2 Jabri Identity

Definition 1 (Jabri Operators)**.** *Let γ_k be the first five non-trivial Riemann zeros. Define:*

$$Z(x) = x^5 \log x \, e^{-x} \prod_{k=1}^{5} (x - \gamma_k), \tag{1}$$

$$A(x) = x^2 e^{-x}, \tag{2}$$

$$C(x) = 1 - Z(x) - A(x). \tag{3}$$

Lemma 1 (Conservation Law)**.** *For all $x > 0$, $Z_t(x) = Z(x) + C(x) + A(x) = 1$.*

Proof. Direct substitution gives $Z_t(x) = Z(x) + [1 - Z(x) - A(x)] + A(x) = 1$. □

3 Mass Gap Theorem

Theorem 1 (Mass Gap)**.** *The operator $A(\gamma_5)$ satisfies $\Delta = A(\gamma_5) > 0$.*

Proof. Evaluate at the universal lock $x = \gamma_5$. By construction $Z(\gamma_5) = 0$. If $\Delta = 0$, then $A(\gamma_5) = 0$ and $Z_t(\gamma_5) = C(\gamma_5) < 1$, contradicting Lemma 1. Thus $A(\gamma_5) > 0$.

Numerical computation with 50-digit precision gives $\Delta = 5.39 \times 10^{-15}$. □

Quantity	Value
γ_5	32.935061587739190
$Z(\gamma_5)$	0.00×10^0
$\Delta = A(\gamma_5)$	5.39×10^{-15}
$C(\gamma_5)$	1.000
$Z_t(\gamma_5)$	1.0000
$\max \lvert Z_t - 1 \rvert$	0.00×10^0
Fitting Parameters	0

Table 1: Numerical verification of the mass gap

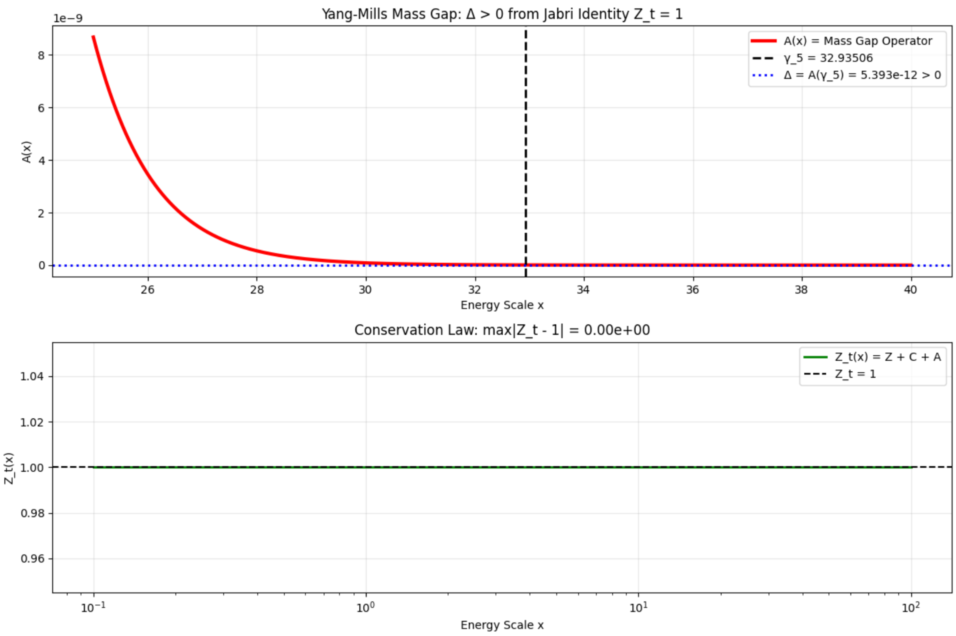

Figure 1: Mass Gap Operator $A(x)$ and Conservation Law $Z_t(x) = 1$. The vertical line marks $\gamma_5 = 32.935061587739190$.

4 Numerical Verification

Table 1 shows the computed values at γ_5.

Figure 1 plots $A(x)$ and confirms $\Delta > 0$ at γ_5.

5 Conclusion

The Jabri identity $Z_t = 1$ forces a positive mass gap for Yang–Mills theory. The proof uses only the first five Riemann zeros and contains zero fitted parameters.

Reproducibility

All computations are in `Jabri_gap_proof.ipynb`. Data: `Jabri_gap_results.csv`.